Jongho Park, Jaeseung Park, Zheyang Xiong, Nayoung Lee, Jaewoong Cho, Samet Oymak, Kangwook Lee, and Dimitris Papailiopoulos. Can mamba learn how to learn? a comparative study on in-context learning tasks, 2024. URL https://arxiv.org/abs/2402.04248.

Keiran Paster, Marco Dos Santos, Zhangir Azerbayev, and Jimmy Ba. Openwebmath: An open dataset of high-quality mathematical web text, 2023.

Roma Patel and Ellie Pavlick. Mapping language models to grounded conceptual spaces. In *International conference on learning representations*, 2022.

Guilherme Penedo, Hynek Kydlíček, Anton Lozhkov, Margaret Mitchell, Colin Raffel, Leandro Von Werra, Thomas Wolf, et al. The fineweb datasets: Decanting the web for the finest text data at scale. *Advances in Neural Information Processing Systems*, 37:30811–30849, 2024.

Alethea Power, Yuri Burda, Harri Edwards, Igor Babuschkin, and Vedant Misra. Grokking: Generalization beyond overfitting on small algorithmic datasets, 2022. URL https://arxiv.org/abs/2201.02177.

Liliang Ren, Yang Liu, Yadong Lu, Yelong Shen, Chen Liang, and Weizhu Chen. Samba: Simple hybrid state space models for efficient unlimited context language modeling, 2025. URL https://arxiv.org/abs/2406.07522.

Olivier Roy and Martin Vetterli. The effective rank: A measure of effective dimensionality. In *2007 15th European signal processing conference*, pages 606–610. IEEE, 2007.

Keisuke Sakaguchi, Ronan Le Bras, Chandra Bhagavatula, and Yejin Choi. Winogrande: An adversarial winograd schema challenge at scale. *Communications of the ACM*, 64(9):99–106, 2021.

Maarten Sap, Hannah Rashkin, Derek Chen, Ronan Le Bras, and Yejin Choi. Social iqa: Commonsense reasoning about social interactions. In *Proceedings of the 2019 conference on empirical methods in natural language processing and the 9th international joint conference on natural language processing (EMNLP-IJCNLP)*, pages 4463–4473, 2019.

J. Schmidhuber. *Making the World Differentiable: On Using Self Supervised Fully Recurrent Neural Networks for Dynamic Reinforcement Learning and Planning in Non-Stationary Environments*. Forschungsberichte Künstliche Intelligenz. Inst. für Informatik, 1990.

Julian Schrittwieser, Ioannis Antonoglou, Thomas Hubert, Karen Simonyan, Laurent Sifre, Simon Schmitt, Arthur Guez, Edward Lockhart, Demis Hassabis, Thore Graepel, et al. Mastering atari, go, chess and shogi by planning with a learned model. *Nature*, 588(7839):604–609, 2020.

Max Schwarzer, Ankesh Anand, Rishab Goel, R Devon Hjelm, Aaron Courville, and Philip Bachman. Data-efficient reinforcement learning with self-predictive representations, 2021. URL https://arxiv.org/abs/2007.05929.

Charlotte Striebel. Sufficient statistics in the optimum control of stochastic systems. *Journal of Mathematical Analysis and Applications*, 12(3):576–592, 1965.

Jianlin Su, Yu Lu, Shengfeng Pan, Ahmed Murtadha, Bo Wen, and Yunfeng Liu. Roformer: Enhanced transformer with rotary position embedding, 2023. URL https://arxiv.org/abs/2104.09864.

Jayakumar Subramanian, Amit Sinha, Raihan Seraj, and Aditya Mahajan. Approximate information state for approximate planning and reinforcement learning in partially observed systems. *Journal of Machine Learning Research*, 23 (12):1–83, 2022.

Richard S Sutton. Dyna, an integrated architecture for learning, planning, and reacting. *ACM Sigart Bulletin*, 2(4): 160–163, 1991.

Yunhao Tang, Zhaohan Daniel Guo, Pierre Harvey Richemond, Bernardo Ávila Pires, Yash Chandak, Rémi Munos, Mark Rowland, Mohammad Gheshlaghi Azar, Charline Le Lan, Clare Lyle, András György, Shantanu Thakoor, Will Dabney, Bilal Piot, Daniele Calandriello, and Michal Valko. Understanding self-predictive learning for reinforcement learning. In *Proceedings of the 40th International Conference on Machine Learning*, ICML'23. JMLR.org, 2023.

Keyon Vafa, Justin Y Chen, Ashesh Rambachan, Jon Kleinberg, and Sendhil Mullainathan. Evaluating the world model implicit in a generative model. *Advances in Neural Information Processing Systems*, 37:26941–26975, 2024.

Keyon Vafa, Peter G Chang, Ashesh Rambachan, and Sendhil Mullainathan. What has a foundation model found? using inductive bias to probe for world models. In *International Conference on Machine Learning*, pages 60727–60747. PMLR, 2025.

Ashish Vaswani, Noam Shazeer, Niki Parmar, Jakob Uszkoreit, Llion Jones, Aidan N Gomez, Ł ukasz Kaiser, and Illia Polosukhin. Attention is all you need. In I. Guyon, U. Von Luxburg, S. Bengio, H. Wallach, R. Fergus, S. Vishwanathan, and R. Garnett, editors, *Advances in Neural Information Processing Systems*, volume 30. Curran Associates, Inc., 2017. URL https://proceedings.neurips.cc/paper_files/paper/2017/file/3f5ee243547dee91fbd053c1c4a845aa-Paper.pdf.

Johannes Welbl, Nelson F Liu, and Matt Gardner. Crowdsourcing multiple choice science questions. In *Proceedings of the 3rd Workshop on Noisy User-generated Text*, pages 94–106, 2017.

Zhaofeng Wu, Linlu Qiu, Alexis Ross, Ekin Akyürek, Boyuan Chen, Bailin Wang, Najoung Kim, Jacob Andreas, and Yoon Kim. Reasoning or reciting? exploring the capabilities and limitations of language models through counterfactual tasks. In Kevin Duh, Helena Gomez, and Steven Bethard, editors, *Proceedings of the 2024 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies (Volume 1: Long Papers)*, pages 1819–1862, Mexico City, Mexico, June 2024. Association for Computational Linguistics. doi: 10.18653/v1/2024.naacl-long.102. URL https://aclanthology.org/2024.naacl-long.102/.

Shunyu Yao, Dian Yu, Jeffrey Zhao, Izhak Shafran, Tom Griffiths, Yuan Cao, and Karthik Narasimhan. Tree of thoughts: Deliberate problem solving with large language models. In A. Oh, T. Naumann, A. Globerson, K. Saenko, M. Hardt, and S. Levine, editors, *Advances in Neural Information Processing Systems*, volume 36, pages 11809–11822. Curran Associates, Inc., 2023. URL https://proceedings.neurips.cc/paper_files/paper/2023/file/271db9922b8d1f4dd7aaef84ed5ac703-Paper-Conference.pdf.

Jiacheng Ye, Jiahui Gao, Shansan Gong, Lin Zheng, Xin Jiang, Zhenguo Li, and Lingpeng Kong. Beyond autoregression: Discrete diffusion for complex reasoning and planning. In *The Thirteenth International Conference on Learning Representations*, 2025. URL https://openreview.net/forum?id=NRYgUzSPZz.

Weirui Ye, Shaohuai Liu, Thanard Kurutach, Pieter Abbeel, and Yang Gao. Mastering atari games with limited data. *Advances in neural information processing systems*, 34:25476–25488, 2021.

Rowan Zellers, Ari Holtzman, Yonatan Bisk, Ali Farhadi, and Yejin Choi. Hellaswag: Can a machine really finish your sentence? In *Proceedings of the 57th annual meeting of the association for computational linguistics*, pages 4791–4800, 2019.

Amy Zhang, Rowan McAllister, Roberto Calandra, Yarin Gal, and Sergey Levine. Learning invariant representations for reinforcement learning without reconstruction. *arXiv preprint arXiv:2006.10742*, 2020.

Guanhua Zhang, Ricardo Dominguez-Olmedo, and Moritz Hardt. Train-before-test harmonizes language model rankings. In *The Fourteenth International Conference on Learning Representations*, 2026. URL https://openreview.net/forum?id=ORv3SAzus1.

Kexin Zhang, Qingsong Wen, Chaoli Zhang, Rongyao Cai, Ming Jin, Yong Liu, James Zhang, Yuxuan Liang, Guansong Pang, Dongjin Song, and Shirui Pan. Self-supervised learning for time series analysis: Taxonomy, progress, and prospects, 2024. URL https://arxiv.org/abs/2306.10125.

A Belief States in Sequence Modeling

Recent work has introduced variants of sequence modeling architectures based on the principle of learning belief states, i.e., BST and JTP. We review these methods here. Let θ denote the parameters of a transformer-based model. Let $\mathbf{h}_{s:t}$ denote the hidden states produced by the transformer encoder for a token sequence $X_{s:t}$, where $s \leq t$. When we use the notation $\mathbf{h}_t$, it is shorthand for $\mathbf{h}_{1:t}$. The model's output head produces a categorical distribution over the token vocabulary conditioned on some hidden state representation, i.e., $p_\theta(\cdot \mid \mathbf{h}_{s:t})$.

Belief State Transformer. The Belief State Transformer (BST; Hu et al. [2025]) learns *compact* belief states by jointly training a next-token predictor and a previous-token predictor across all possible prefix–suffix decompositions of a sequence, including cases where either the prefix or suffix is empty. Concretely, given a prefix $X_{1:t}$ and a suffix $X_{t+k:T}$ with $k \geq 1$, BST aims to minimize the cross-entropy loss

$$\mathcal{L}_{\text{BST}}(\theta) = \mathbb{E}_{t<T}\Big[-\log \underbrace{p_\theta(X_{t+1} \mid \mathbf{h}_{1:t}, \mathbf{h}_{t+k:T})}_{\text{next-token prediction}} - \log \underbrace{p_\theta(X_{t+k-1} \mid \mathbf{h}_{1:t}, \mathbf{h}_{t+k:T})}_{\text{previous-token prediction}} \Big], \tag{6}$$

where $\mathbf{h}_{1:t}$ and $\mathbf{h}_{t+k:T}$ are produced by separate transformers. This bidirectional training shapes the hidden representations of the BST into belief states.

Joint Multi-Token Prediction. Joint multi-token prediction (JTP; Ahn et al. [2025]) aims to learn the *joint* distribution over the next $d+1$ tokens conditioned on $\mathbf{h}_t$, where d is the multi-step prediction horizon beyond the next token. Specifically, JTP minimizes the loss

$$\mathcal{L}_{\text{JTP}}(\theta; d, \lambda_{\text{MTP}}) = \mathbb{E}_{t<T}\Big[-\log \underbrace{p_\theta(X_{t+1} \mid \mathbf{h}_t)}_{\text{next-token prediction}} - \lambda_{\text{MTP}} \cdot \frac{1}{d} \sum_{i=1}^{d} \log \underbrace{p_\theta(X_{t+i+1} \mid \text{Fetch}(\mathbf{h}_t, X_{t+1:t+i}))}_{\text{joint multi-token prediction}} \Big], \tag{7}$$

where an additional module $\text{Fetch}(\mathbf{h}_t, X_{t+1:t+i})$ is used to create an embedding combining the teacher-forced tokens $X_{t+1:t+i}$ with $\mathbf{h}_t$ and $\lambda_{\text{MTP}} > 0$ balances next- and multi-token prediction losses. Although Ahn et al. [2025] suggest that their method learn "short-horizon belief states", they do not formally define the conditions under which this occurs. To understand how JTP learns belief states, we start by defining a *k-observable* system.

Definition A.1 (k-observability for sequences)**.** A system is k-observable if for any two sequences $H = X_{1:t}$ and $H' = X_{1:j}$ that induce the same joint distribution over the next-k tokens, i.e., $\mathbb{P}(X_{t+1:t+k} \mid H) = \mathbb{P}(X_{t+1:t+k} \mid H')$, it follows that their *full-horizon* conditionals match:

$$\mathbb{P}(X_{t+1:T} \mid H) = \mathbb{P}(X_{t+1:T} \mid H'). \tag{8}$$

In other words, the distribution of all future observations is determined by the distribution of the next k observations.

Proposition A.2 (JTP forms belief states in k-observable systems)**.** *Assume the system is k-observable and let $k = d+1$. Suppose the joint multi-token prediction model recovers the true joint next-k conditional, i.e. $p_\theta(X_{t+1} \mid \mathbf{h}_t)p_\theta(X_{t+2} \mid \mathbf{h}_t, X_{t+1}) \dots p_\theta(X_{t+k} \mid \mathbf{h}_t, X_{t+1:t+k-1}) = \mathbb{P}(X_{t+1:t+k} \mid X_{1:t})$ a.s. for all t, then $\mathbf{h}_t$ is a belief state for $X_{1:t}$.*

Proof. By k-observability (Theorem A.1), there exists a measurable map G taking the joint next-k conditional distribution to the full-horizon conditional:

$$\mathbb{P}(X_{t+1:T} \mid X_{1:t}) = G\Big(\mathbb{P}(X_{t+1:t+k} \mid X_{1:t})\Big).$$

By the premise that JTP recovers the true next-k joint, conditioning on $\mathbf{h}_T$ is equivalent to conditioning on $X_{1:t}$ for all bounded measurable functionals of the future. Hence $\mathbf{h}_t$ is a belief state for all $t < T$. □

Intuitively, if all possible futures can be distinguished by the next k tokens, then a JTP model that accurately predicts the joint next-k distribution would encode all information necessary to distinguish future trajectories. Note that both next-token prediction and multi-token prediction do not guarantee belief state representations (see Hu et al. [2025]).

B Formal Proof of Theorem 3.2

The proof follows the intuition illustrated below. Optimizing for next-token and transition consistency (Equations 1 and 2) ensures the existence of measurable maps p_θ and p_ψ that allow recursive decoding of future tokens:

$$\mathbf{h}_t \xrightarrow[\text{decode token}]{p_\theta} X_{t+1} \xrightarrow[\text{update state}]{p_\psi} \mathbf{h}_{t+1} \xrightarrow[\text{decode token}]{p_\theta} X_{t+2} \xrightarrow[\text{update state}]{p_\psi} \mathbf{h}_{t+2} \cdots \xrightarrow{p_\theta} X_T.$$

Proof. A formal proof proceeds by backward induction on t. For the base case $t = T-1$, the claim follows directly from Equation 1, since $\mathbf{h}_{T-1}$ suffices to predict X_T.

Now assume $\mathbf{h}_{k+1}$ is a belief state for $X_{1:k+1}$. By Theorem 3.1, this implies that $X_{k+2:T}$ is conditionally independent of $X_{1:k+1}$ given $\mathbf{h}_{k+1}$. We will show that $\mathbf{h}_k$ is also a belief state for $X_{1:k}$. From $\mathbf{h}_k$, one can generate

$$X_{k+1} \sim p_\theta(\cdot \mid \mathbf{h}_k),$$
$$\mathbf{h}_{k+1} \sim p_\psi(\cdot \mid \mathbf{h}_k, X_{k+1}).$$

By next-token and transition consistency (Equations 1 and 2), we have

$$\begin{aligned}
\mathbb{P}(X_{k+1:T} \mid \mathbf{h}_k) &= \mathbb{P}(X_{k+2:T} \mid X_{k+1}, \mathbf{h}_k)\, \underbrace{\mathbb{P}(X_{k+1} \mid \mathbf{h}_k)}_{\text{Equation 1}} \\
&= \mathbb{P}(X_{k+2:T} \mid X_{k+1}, \mathbf{h}_k)\, \mathbb{P}(X_{k+1} \mid X_{1:k}) \\
&= \left[\int \mathbb{P}(X_{k+2:T},\ \mathbf{h}_{k+1} \mid X_{k+1}, \mathbf{h}_k)\, d\mathbf{h}_{k+1}\right] \mathbb{P}(X_{k+1} \mid X_{1:k}) \\
&= \left[\int \underbrace{\mathbb{P}(X_{k+2:T} \mid \mathbf{h}_{k+1}, X_{k+1}, \mathbf{h}_k)\, \mathbb{P}(\mathbf{h}_{k+1} \mid X_{k+1}, \mathbf{h}_k)}_{\text{Equation 2}}\, d\mathbf{h}_{k+1}\right] \mathbb{P}(X_{k+1} \mid X_{1:k}) \\
&= \left[\int \underbrace{\mathbb{P}(X_{k+2:T} \mid \mathbf{h}_{k+1})}_{\text{Induction hypothesis}}\, \mathbb{P}(\mathbf{h}_{k+1} \mid X_{1:k+1})\, d\mathbf{h}_{k+1}\right] \mathbb{P}(X_{k+1} \mid X_{1:k}) \\
&= \mathbb{P}(X_{k+2:T} \mid X_{1:k+1})\, \mathbb{P}(X_{k+1} \mid X_{1:k}) = \mathbb{P}(X_{k+1:T} \mid X_{1:k}).
\end{aligned}$$

This proves that $\mathbf{h}_k$ is also a belief state. □

C More Details on NextLat Implementation

We parameterize the latent transition model p_ψ with a three-layer MLP using GELU [Hendrycks and Gimpel, 2023] activations. The latent transition model takes as input the layer-normalized [Ba et al., 2016] concatenation of the current hidden state $\mathbf{h}_t$ and next-token embedding X_{t+1}, and outputs a *delta* update applied via residual connection:

$$\hat{\mathbf{h}}_{t+1} = p_\psi(\mathbf{h}_t, X_{t+1}) = f_\psi(\mathbf{h}_t, X_{t+1}) + \mathbf{h}_t \tag{9}$$

where $f_\psi(\cdot)$ predicts the modification to $\mathbf{h}_t$ (see Figure 11). This paper aims to demonstrate that NextLat yields significant performance gains even with a **simple MLP latent transition**. We foresee even better performances with more sophiscated latent transition model architectures, but we leave that exploration to future work.

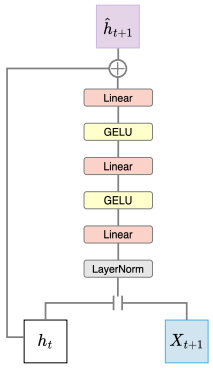


Figure 11: Illustration of the latent transition model p_ψ.

Following standard training, we mask token-level losses (i.e., $\mathcal{L}_{\text{next-token}}$ and $\mathcal{L}_{\text{KL}}$) corresponding to context or prompt tokens. However, we do not apply masking for $\mathcal{L}_{\text{next-h}}$ on context tokens, ensuring that belief state representations develop even during context processing. When using sequence packing (e.g., for Manhattan, TinyStories, and FineWeb-Edu), we mask $\mathcal{L}_{\text{KL}}$ and $\mathcal{L}_{\text{next-h}}$ terms that cross document boundaries.